# Mathematical Methods in Image Processing (MA3221)

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## 1 Instructions

- Total marks for the project is 30.
- Each group is divided into 5 members. The contribution of each team member should be precisely explained during the project presentation.
- A project report should be prepared highlighting the novelty, methodology, key observations, summary/conclusions, future scope and references.

## 2 List of Projects

| Group Number | Project | |
|---|---|---|
| 1 | TNRD with $u_t = u_{t-1} - \left(\sum_{i=1}^{N_k} \bar{k}_i^t * (\frac{u_\sigma}{M} * \phi_i^t(k_i^t * u_{(t-1)p})) + \lambda(\frac{u-f}{u^2+\epsilon})\right)$<br>for multiplicative gamma noise removal ($L = 1 \& L = 10$).<br>Fix $\phi_i$ and learn other parameters stage-wise.<br>Compare the results with the PDE model<br>Conduct ablation study | https://lin |
| 2 | TNRD with $u_t = u_{t-1} - \left(\sum_{i=1}^{N_k} \bar{k}_i^t * (\frac{u_\sigma}{M} * \phi_i^t(k_i^t * u_{(t-1)p})) + \lambda(\frac{u-f}{u^2+\epsilon})\right)$<br>for multiplicative gamma noise removal ($L = 1 \& L = 10$).<br>Fix $k_i$ and learn other parameters stage-wise.<br>Conduct ablation study.<br>Compare the results with the PDE model | https://lin |
| 3 | TNRD with $u_t = u_{t-1} - \left(\sum_{i=1}^{N_k} \bar{k}_i^t * (\frac{u_\sigma}{M} * \phi_i^t(k_i^t * u_{(t-1)p})) + \lambda(\frac{u-f}{u^2+\epsilon})\right)$<br>for multiplicative gamma noise removal ($L = 1 \& L = 10$).<br>Learn all the parameters $\phi_i, k_i, \lambda$ stage-wise.<br>Compare the results with the PDE model | https://lin |
| 4 | Develop a learnable architecture for $u_{tt} + \gamma u_t = div(c(\|\nabla u\|)\nabla u)$<br>for additive Gaussian noise<br>removal and compare with TNRD, PDE models for a chosen noise level.<br>Fix $\phi_i, \gamma$ and learn other parameters stage-wise. | htt |
| 5 | Develop a learnable architecture for $u_{tt} + \gamma u_t = div(c(\|\nabla u\|)\nabla u)$<br>for additive Gaussian noise<br>removal and compare with TNRD, PDE models for a chosen noise level.<br>Conduct ablation study.<br>Fix $k_i, \gamma$ and learn other parameters stage-wise. | htt |
| 6 | Develop a learnable architecture for $u_{tt} + \gamma u_t = div(c(\|\nabla u\|)\nabla u)$<br>for additive Gaussian noise<br>removal and compare with TNRD, PDE models for a chosen noise level.<br>Fix $\gamma$ and learn all the parameters $\phi_i, k_i, \lambda$ stage-wise.<br>Conduct ablation study. | htt |
| 7 | Develop an architecture for $u_{tt} + \gamma u_t = div(g(u_\sigma)c(\|\nabla u\|)\nabla u)$<br>for multiplicative gamma noise removal<br>and compare with the underlying PDE model for $L = 1 \& L = 10$.<br>Fix $\phi_i, \gamma$ and learn other parameters stage-wise.<br>Conduct ablation study. | htt |
| 8 | Develop an architecture for $u_{tt} + \gamma u_t = div(g(u_\sigma)c(\|\nabla u\|)\nabla u)$<br>for multiplicative gamma noise removal<br>and compare with the underlying PDE model for $L = 1 \& L = 10$.<br>Fix $k_i, \gamma$ and learn other parameters stage-wise.<br>Conduct ablation study. | htt |



Table 1: Projects list

| Group Number | Project | |
|---|---|---|
| 9 | Develop an architecture for $u_{tt} + \gamma u_t = div(g(u_\sigma)c(\lvert\nabla u\rvert)\nabla u)$<br>for multiplicative gamma noise removal<br>and compare with the underlying PDE model for $L = 1 \& L = 10$.<br>Fix $\gamma$ and learn all the parameters $\phi_i, k_i, \lambda$ stage-wise.<br>Conduct ablation study. | https://epu |
| 10 | Develop an architecture for $u_{tt} + \gamma u_t = div(c(\lvert\nabla u\rvert)\nabla u)$<br>for Poisson noise<br>removal (use same reaction term of referred paper)<br>and compare with the referred paper. Fix $\gamma$ and learn other parameters. | https://ie |
| 11 | Develop an architecture for $u_t = div(g(u_\sigma)c(\lvert\nabla u\rvert)\nabla u)$<br>for Poisson noise removal<br>(use same reaction term of referred paper)<br>and compare with the referred model any two noise levels. | https://ie |
| 12 | Develop an architecture for $u_t^\alpha = div(c(\lvert\nabla u\rvert)\nabla u)$<br>for additive noise<br>removal (use same reaction term of referred paper)<br>and compare with the referred paper.<br>Fix $\alpha$ and learn other parameters.<br>Conduct ablation study. | |
| 13 | Develop an architecture for $u_t^\alpha = div(g(u_\sigma)c(\lvert\nabla u\rvert)\nabla u)$<br>for multiplicative noise removal<br>(use same reaction term of referred paper)<br>and compare with the referred model any two noise levels.<br>Fix $\alpha$ and learn other parameters | https://ie |
| 14 | Develop an architecture for $u_t^\alpha = div(c(\lvert\nabla u\rvert)\nabla u)$<br>for additive noise<br>removal (use same reaction term of referred paper)<br>and compare with the referred paper.<br>Train all the parameters including $\alpha$. | |

Table 2: Projects list